

An immersive simulator for improving chemistry learning efficiency

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ABSTRACT

This paper designed a virtual simulator for chemistry education and computer-mediated hands-on exercises. We compared the participants' performances in hands-on chemistry tasks with or without score-keeping and timer and used questionnaires to measure their learning performance and perceived cognitive workload. The experimental results indicate participants' preferences and attitudes toward efficiency and safety. Most participants reported that the learning simulator could improve learning efficiency and a sense of safety. Our findings shed light on the quality and learning performance of theory knowledge and operational skills for chemistry education.

Index Terms: Applied computing—Education—Interactive learning environments Human-centered computing—HCI—Interaction paradigms—Virtual reality

1 INTRODUCTION

With the development of virtual reality (VR), virtual environments are gaining popularity among educational applications in the education of VR applications are popular and active in diversified fields and industries [1]. Due to the disadvantages of current experiments in a physical lab, many scholars use VR technology to improve the shortcomings of offline experiments. Although various VR simulators have been built and shown to increase students' learning efficiency and experimental safety via VR, only some have addressed the influence of the heartbeat rate of stressed environments on students' learning efficiency in VR. Therefore, VR education studies must determine how to assess students' success in VR chemical experiments and motivate them to complete operations more efficiently. To address this issue, in this paper, our primary research question is whether a stressed atmosphere in VR generated from specific occasions can improve the learning efficiency of operation skills.

2 METHOD

We use Unity 3D¹ to design a VR chemistry lab with the examination system in Oculus Quest 2 to validate whether mental stress has an impact on students' learning efficiency. When we activate the examination system in the VR environment, it will assess the operational performance of the students and provide them with a score evaluation and overall evaluation (excellent, good, unqualified). In addition, an E4 wristband² was used to monitor the user's heart

rate and experiment duration. The heartbeat rate indicator measured by the E4 wristband can demonstrate the user's physiological states in the VR environment [2]. Participants were randomly divided into two groups and asked to complete two experiments (one of the experiments needed to open the examination system, and the other experiment did not). Each experiment necessitates the completion of two solid oxide fuel cell-related tasks (self-propagating high-temperature synthesis (E1) and electrode slice fabrication (E2)). To avoid the learning effect and unexpected errors, the order of the experiments was randomized.

3 VIRTUAL SIMULATOR DEVELOPMENT

This application consists of four parts: *video presentation*, *integrated learning center*, *lab hub*, and *other learning tools*. Fig. 1 illustrates the structure. The *video presentation center* is about basic VR knowledge and how to manage the VR operation quickly. The *integrated learning center* provides chemistry knowledge materials with some panels, and students could customize the system to different color styles, even dark modes. *Lab hub* is a simulated chemistry environment based on the real chemistry laboratory where students can perform chemistry experiments. Finally, the *other learning tools* offer a choice-question test panel for students. The advantages of our VR simulator as illustrated: Functional Diversity, Laboratory Safety, and Hazardous Experience (Fig. 2) and Student Performance Assessment (Fig. 3).



Figure 1: Four main parts in the learning environment: A) Video presentation center; B) Integrated learning centre; C) Lab hub; D) Other learning tools (a multiple-choice answering module).

4 USER STUDY

Thirty-nine participants from the local university were recruited to complete this study (26 male and 13 female, $Age_{mean} = 22$ years, all right-handed). We employed Qualtrics to design the questionnaires, including basic information (Q1), first NASA Task Load Index (6 questions) (Q2), second NASA Task Load Index (6 questions) (Q3), and Efficiency Feedback (2 questions) (Q4). The experimental procedure for the participants in this study is as follows: A) Wear the E4

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¹<https://unity.com>

²<https://www.empatica.com/research/e4/>

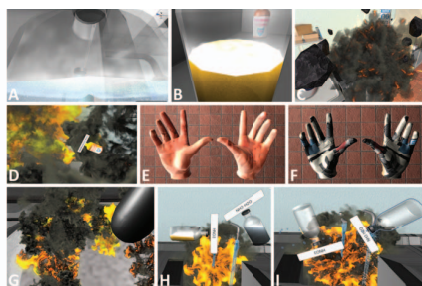


Figure 2: Five different VR chemistry hazard displays: A) Volatile irritating gas on the surface of concentrated ammonia water; B) Volatile toxic gases on the surface of concentrated nitric acid; C) Laboratory oven explosion effect; D) The chemical solution and the water in the sink mixed and exploded; E) the virtual hands in the normal state; F) The virtual hands after being corroded by the chemical solution; G) In self-propagating high-temperature synthesis, excessive stirring time leads to solution explosion and sputtering; H and I) Combustion of hazardous chemicals.

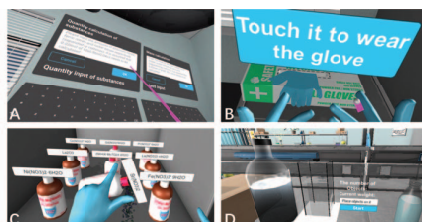


Figure 3: User interaction with virtual objects: A) Calculating the quantity and mass of objects; B) Wear the glove; C) Preparation of solution; D) Weigh objects with a balance.

wristband; B) Wear the VR headset; C) Finish two experiments in the VR simulator in condition one; D) Fill in the questionnaire(Q1 and Q2) to report the cognitive workload; E) Finish two experiments in the VR simulator in condition two; F) Fill in the questionnaire(Q3 and Q4) to compare the immersive learning efficiency with and without an exam. Conditions 1 and 2 were randomly assigned to the exam).

Table 1: The majority of samples (N = 39) report significantly in NASA-TLX with extra aspects (time, heartbeat rate, and learning efficiency).

Aspect	Positive Number	Percentage	χ^2	P-value	ϕ
Mental Demand	33	84.62%	18.692	<.01**	0.692
Physical Demand	31	79.49%	13.564	<.01**	0.59
Temporal Demand	30	76.92%	11.308	<.01**	0.538
Performance	31	79.49%	13.564	<.01**	0.59
Effort	32	82.05%	16.026	<.01**	0.641
Frustration	28	71.79%	7.41	<.01**	0.436
Time	16	41.03%	1.256	0.26	0.18
Heart Beat Rate	32	82.05%	16.026	<.01**	0.641
Learning Efficiency	37	94.87%	31.41	<.01**	0.897

5 RESULT AND DISCUSSION

The ultimate data results from the analysis are presented in Table 1. The “Positive Number” in this table means the score in the Exam part greater than the No-Exam. Regarding the NASA-TLX questionnaire, we use the χ^2 to measure the cognitive workload imposed

on participants from 6 aspects of the learning simulator, and results indicate that the majority of the user experienced increased cognitive workload ($p < .01$). The examination scene has enhanced their cognitive load, and the participant’s heart rate is greater than the one in the situation without exam ($p < .01$). This is also in line with the NASA-TLX questionnaire results of users. Most individuals think they need more mental, physical, and temporal demands to finish the VR chemistry experiment in this study, and the performance, effort, and frustration attributes are positive as well. Finally, 94.87% of participants report that the stressed atmosphere with the exam could improve the learning efficiency in the virtual learning simulator.

However, the result indicated there is no significant time difference for the task with an exam and without an exam ($p = .26$). There is no sufficient evidence to ensure that a higher cognitive load will make them quickly finish experiments. The reasons might be the following. The virtual laboratory environment, the emotion of participants, and the familiarity of chemistry majors could impact the task time. For example, participants who have previously learned the self-propagating high-temperature synthesis may spend less time finishing the examination in VR. Then, it is possible that participants can screw up with the tasks and still get a shorter completion time. Additionally, users with poor tolerance to the VR environment may experience subjective discomfort in the virtual scene, which in turn may adversely affect physiological indicators measured in the experiment (such as heartbeat rate), and then affect the total completion time [3].

According to the participants’ feedback after the experiment, a majority of them agreed that the examination system could improve their learning attitude because they were worried that poor performance might influence the exam score. Focus on the relevance of learning efficiency and experiment time, most of them regard the exam system plays more roles in enhancing the seriousness and quality of finishing an experiment instead of the task completion time. They will prioritize how to complete each operation of the experiment carefully instead of completing the experiment faster in virtual exam occasions.

6 CONCLUSION

This paper measured the user’s heart rate and task completion time in various VR environments by establishing a highly realistic laboratory and combining the results with questionnaire surveys. The user evaluation investigates the effect of heart rate variability on the efficiency of students’ practical manipulation skill acquisition in virtual reality. It is essential to note that by creating a serious and stressful environment (the VR exam environment), students can improve their experimental operation skills and learning efficiency. In the meantime, the examination system in the VR environment can assist students in completing tasks with greater care, thereby reducing experimental errors and safety accidents. In the future, our research may assist other disciplines in determining how to utilize virtual reality educational applications to evaluate student performance in experiments.

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